THARUNKUMAR K

FULL STACK DEVELOPER | AI APPLICATION DEVELOPER | SDE ASPIRANT

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PROFESSIONAL SUMMARY

Full Stack Developer experienced in building scalable web applications, AI-powered systems, and backend services using React, Next.js, Node.js, Java, and Python. Interned at Neura Global and Infosys Springboard delivering production-grade software and AI integrations.

TECHNICAL SKILLS

Languages:	TypeScript, JavaScript, Python, Java, C, C++, SQL (PostgreSQL, MySQL)
Frontend:	React 19, Next.js, Vite, Tailwind CSS, Monaco Editor, HTML5, CSS3
Backend & Systems:	Node.js, Express.js, Spring Boot, REST APIs, Socket.IO, WebRTC, Prisma ORM
Databases & Cloud:	PostgreSQL, Supabase, MongoDB, SQLite, Docker, Vercel, Render
AI & Developer Tools:	Google Gemini AI API, Ollama (Local RAG), Git, GitHub, GitHub Actions, Cloudinary

EDUCATION

Bachelor of Technology (B.Tech) in Information Technology — VSB Engineering College, Karur, TN **2024 – 2028**

Cumulative GPA: **8.5 / 10.0**

PROFESSIONAL EXPERIENCE

Full-Stack Development Intern — Neura Global **06/2026 – 07/2026**

- **Campus Recruitment Platform:** Engineered **CampusConnect**, managing 5-stage recruitment funnels for **500+ student profiles** with 4-role RBAC security using React 19, TypeScript, Node.js, Express, and Supabase.
- **AI Semantic Screening:** Integrated **Google Gemini AI** for resume screening and **Socket.IO** real-time updates, cutting manual roster auditing by **70–80%** with zero scheduling clashes and Cloudinary document storage.

Software Development Intern — Infosys Springboard **11/2025 – 01/2026**

- **Enterprise Recognition System:** Developed **StarWall** peer recognition and analytics dashboard using Java 17, Spring Boot, modular microservices, and test-driven development (TDD) adhering to agile Git workflows.

FLAGSHIP PROJECTS

VSBEC IT Vault — Academic Task Manager & Compiler Platform | *Top 50 SIH Nominee* **2026**

React 19, *TypeScript*, *Tailwind CSS*, *Node.js*, *Express*, *PostgreSQL*, *Monaco Editor*, *Docker* `[Demo]` | `[GitHub]`

- **Campus Production Adoption:** Deployed and actively used daily by **365+ enrolled students** across 6 sections for assignment submissions, proof verification, and automated LeetCode/GitHub sync.
- **Isolated Compiler Sandbox:** Built multi-language execution runtime (**C, C++, Java 17, Python 3**) with isolated child-process jails, timeout traps, server-side test evaluation, and a 3-tier approval hierarchy.

GOAT Code Editor (GOAT-CE) — Real-Time Collaborative Cloud IDE | *1st Place National Winner* **2026**

React 19.2, *TypeScript*, *Pure WebRTC*, *Socket.IO 4.8*, *Monaco Editor Core*, *PostgreSQL*, *Piston API* `[Demo]` | `[GitHub]`

- **Zero-SDK WebRTC Voice Mesh:** Built custom P2P audio mesh handling SDP handshakes, ICE candidates, and hardware echo cancellation without paid 3rd-party SDKs for ultra-low latency collaboration.
- **Sub-Pixel Diff Streaming:** Implemented Socket.IO pipeline streaming operational transformations, remote cursor presence, and live selection across Monaco editor. Won **1st Place (₹5,000)** at Code Thugs 2K26.

Smart Helmet IoT — Rider Safety & Accident Prevention System | *Top 50 SIH Nominee* **2025**

Embedded C/C++, *Arduino Nano*, *RF 433MHz*, *MQ-3 Alcohol Sensor*, *IR Proximity*, *Eye-Blink Sensor*, *SPDT Relay* `[GitHub]`

- **Dual-Unit Ignition Interlock:** Engineered embedded safety system linking helmet transmitter and bike receiver over RF 433MHz; enforces optical IR wear detection and MQ-3 breath alcohol sensing with **<500ms automated ignition cut-off**.
- **Drowsiness Alert & Fail-Safe Watchdog:** Integrated IR eye-blink eyelid sensor for real-time micro-sleep alerts, implementing a **5s RF fail-safe watchdog** that automatically isolates vehicle ignition upon signal loss or tampering.

AURA — AI Unified Retrieval Assistant (Air-Gapped Local RAG) **2026**

Java 17, *Spring Boot 3*, *React*, *Vite*, *Electron*, *SQLite*, *ChromaDB*, *Ollama (LLaMA 3)*, *WebSockets* `[GitHub]`

- **100% Offline RAG Pipeline:** Built enterprise desktop assistant executing local LLMs (Ollama) with loopback network isolation, PDF vector search, and real-time WebSocket token streaming with zero API costs.
- **Cross-Format Intelligence:** Implemented local document parsing, semantic search over enterprise archives, and local speech-to-text processing packaged into an Electron desktop bundle.

HONORS & ACHIEVEMENTS

- **1st Place National Winner** — *Code Thugs 2K26* (National Hackathon Champion for GOAT Collaborative IDE; ₹5,000 prize).
- **National Finalist** — *Odoo × NMIT Bangalore Hackathon 2026* (Qualified for 24-hr Grand Finale at NMIT Bengaluru).
- **Top 50 / 300+ Teams** — *Smart India Hackathon (SIH 2026 & SIH 2025)* (Portal Nominee for VSBEC IT Vault and Smart Helmet IoT).
- **3rd Prize Winner** — *8-Hour AI/ML Hackathon 2026* (Awarded ChatGPT Go Subscription by Dept of AI&ML for AI innovation).
- **National Finalist** — *India Innovates 2026* (National Innovation Showcase).

LEADERSHIP & RESPONSIBILITIES

- **IT Department Student Coordinator (2024 – Present):** Coordinating student initiatives and symposium events for 150+ attendees.
- **Smart India Hackathon (SIH) Department Coordinator:** Mentoring student teams and coordinating department-wide submissions.
- **GeeksforGeeks Campus Mantri (Jan – Jul 2026):** Organized competitive programming bootcamps for 200+ students.

VERIFIED CERTIFICATIONS

AI & Cloud:	Google Cloud: Responsible AI, Intro to LLMs, Vertex AI Prompt Design; Tata GenAI Data Analytics (Forage)
Software & IoT:	Infosys Springboard Internship Certificate; Cisco Intro to IoT; TCS iON Career Edge; Salesforce Admin